

Module-2

1) Fraud Transaction Detection (Linear Search)

Logic:-

- Take as input of transactions Id and Target
- Then match the target with Transaction Id by One-by-One
- If match then print fraud.

Code:-

```
int main () {  
    int n, target;  
    cin >> n;  
    int arr[n];  
    for (int i = 0; i < n; i++)  
        cin >> arr[i];  
    cin >> target;  
    bool found = false;  
    for (int i = 0; i < n; i++) {  
        if (arr[i] == target) {  
            found = true;  
            break;  
        }  
    }  
    if (found)  
        cout << "fraud transaction found";  
    else  
        cout << "Transaction not found";  
    return 0;  
}
```

Code.cpp >  code.cpp >  main()

```
1  #include <iostream>
2  using namespace std;
3
4  int main() {
5      int n, target;
6      cin >> n;
7
8      int arr[n];
9
10     for(int i = 0; i < n; i++)
11         cin >> arr[i];
12
13     cin >> target;
14
15     bool found = false;
16
17     for(int i = 0; i < n; i++) {
18         if(arr[i] == target) {
19             found = true;
20             break;
21         }
22     }
23
24     if(found)
25         cout << "Fraud Transaction Found";
26     else
27         cout << "Transaction Not Found";
28
29     return 0;
30 }
```

PROBLEMS OUTPUT DEBUG CONSOLE TERMINAL PORTS

akashrathor@Akashs-MacBook-Air code.cpp % clear

5

123 456 789 234 578

456

Fraud Transaction Found

akashrathor@Akashs-MacBook-Air code.cpp %

2) Search Lag Search (Binary Search)

Logic :-

- A array is already Sorted
- find middle element.
- If $\text{target} < \text{middle}$, Search left half
- If $\text{target} > \text{middle}$, Search right half
- Continue untill found or range ends.

Code :-

```
int main () {
    int n, target;
    cin >> n;
    int arr[n];
    for (int i = 0; i < n; i++)
        cin >> arr[i];
    cin >> target;
    int low = 0, high = n - 1;
    bool found = false;
    while (low <= high) {
        int mid = (low + high) / 2;
        if (arr[mid] == target) {
            found = true;
            break;
        }
        else if (target < arr[mid])
            high = mid - 1;
        else low = mid + 1;
    }
    if (found)
        cout << "lag found";
    else
        cout << "lag Not found";
    return 0;
}
```

code.cpp ×

Code.cpp > code.cpp > main()

```
1  #include <iostream>
2  using namespace std;
3
4  int main() {
5      int n, target;
6      cin >> n;
7
8      int arr[n];
9
10     for(int i = 0; i < n; i++)
11         cin >> arr[i];
12
13     cin >> target;
14
15     int low = 0, high = n - 1;
16     bool found = false;
17
18     while(low <= high) {
19         int mid = (low + high) / 2;
20
21         if(arr[mid] == target) {
22             found = true;
23             break;
24         }
25         else if(target < arr[mid])
26             high = mid - 1;
27         else
28             low = mid + 1;
29     }
30
31     if(found)
32         cout << "Log Found";
33     else
34         cout << "Log Not Found";
35
36     return 0;
37 }
```

PROBLEMS OUTPUT DEBUG CONSOLE TERMINAL PORTS

akashrathore@Akashs-MacBook-Air code.cpp % clear

6

4 6 8 10 12 14

10

Log Found

○ akashrathore@Akashs-MacBook-Air code.cpp %

zsh

Code code....

3) Sort Employee Salaries (Bubble Sort).

Logic:-

- Compare adjacent elements
- Swap if left element is greater.
- Largest element moves to end in each pass.

Code:-

```
int main () {  
    int n;  
    cin >> n;  
    int arr[n];  
    for (int i = 0; i < n; i++)  
        cin >> arr[i];  
    for (int i = 0; i < n - 1; i++) {  
        for (int j = 0; j < n - i - 1; j++) {  
            if (arr[j] > arr[j + 1]) {  
                int temp = arr[j];  
                arr[j] = arr[j + 1];  
                arr[j + 1] = temp;  
            }  
        }  
    }  
    for (int i = 0; i < n; i++)  
        cout << arr[i] << " ";  
    return 0;  
}
```

Code.cpp >  code.cpp >  main()

```
1  #include <iostream>
2  using namespace std;
3
4  int main() {
5      int n;
6      cin >> n;
7
8      int arr[n];
9
10     for(int i = 0; i < n; i++)
11         cin >> arr[i];
12
13     for(int i = 0; i < n - 1; i++) {
14         for(int j = 0; j < n - i - 1; j++) {
15             if(arr[j] > arr[j + 1]) {
16                 int temp = arr[j];
17                 arr[j] = arr[j + 1];
18                 arr[j + 1] = temp;
19             }
20         }
21     }
22
23     for(int i = 0; i < n; i++)
24         cout << arr[i] << " ";
25
26     return 0;
27 }
```

PROBLEMS OUTPUT DEBUG CONSOLE TERMINAL PORTS

akashrathor@Akashs-MacBook-Air code.cpp % clear

5

1200 4300 5600 4320 7650

1200 4300 4320 5600 7650

○ akashrathor@Akashs-MacBook-Air code.cpp %

 zsh

 Code code....

> Sort website Response time (Insertion Sort)

Logic :-

- Assume first element is sorted
- pick next element
- Insert it into correct position in sorted part.

Code:-

```
int main() {  
    int n;  
    cin >> n;  
    int arr[n];  
    for(int i=0; i<n; i++)  
        cin >> arr[i];  
    for(int i=1; i<n; i++) {  
        int key = arr[i];  
        int j = i-1;  
        while (j >= 0 && arr[j] > key) {  
            arr[j+1] = arr[j];  
            j--;  
        }  
        arr[j+1] = key;  
    }  
    for(int i=0; i<n; i++)  
        cout << arr[i] << " ";  
    return 0;  
}
```

code.cpp

Code.cpp > code.cpp > main()

```
2  using namespace std;
3
4  int main() {
5      int n;
6      cin >> n;
7
8      int arr[n];
9
10     for(int i = 0; i < n; i++)
11         cin >> arr[i];
12
13     for(int i = 1; i < n; i++) {
14         int key = arr[i];
15         int j = i - 1;
16
17         while(j >= 0 && arr[j] > key) {
18             arr[j + 1] = arr[j];
19             j--;
20         }
21
22         arr[j + 1] = key;
23     }
24
25     for(int i = 0; i < n; i++)
26         cout << arr[i] << " ";
27
28     return 0;
29 }
```

PROBLEMS

OUTPUT

DEBUG CONSOLE

TERMINAL

PORTS

akashrathor@Akashs-MacBook-Air code.cpp % clear

5

245 678 987 543 678

245 543 678 678 987

akashrathor@Akashs-MacBook-Air code.cpp %

zsh

Code code....

5. Priorities Bug Reports (Selection Sort)

Logic:-

- Find Smallest element.
- Swap it with first position
- Repeat for remaining array

Code:-

```
int main () {  
    int n;  
    cin >> n;  
    int arr[n];  
    for (int i = 0; i < n; i++)  
        cin >> arr[i];  
  
    for (int i = 0; i < n - 1; i++) {  
        int minIndex = i;  
        for (int j = i + 1; j < n; j++) {  
            if (arr[j] < arr[minIndex])  
                minIndex = j;  
        }  
        int temp = arr[i];  
        arr[i] = arr[minIndex];  
        arr[minIndex] = temp;  
    }  
  
    for (int i = 0; i < n; i++)  
        cout << arr[i] << " ";  
  
    return 0;  
}
```

code.cpp ×

Code.cpp > code.cpp > main()

```
4 int main() {
5     int n;
6     cin >> n;
7
8     int arr[n];
9
10    for(int i = 0; i < n; i++)
11        cin >> arr[i];
12
13    for(int i = 0; i < n - 1; i++) {
14
15        int minIndex = i;
16
17        for(int j = i + 1; j < n; j++) {
18            if(arr[j] < arr[minIndex])
19                minIndex = j;
20        }
21
22        int temp = arr[i];
23        arr[i] = arr[minIndex];
24        arr[minIndex] = temp;
25    }
26
27    for(int i = 0; i < n; i++)
28        cout << arr[i] << " ";
29
30    return 0;
31 }
```

PROBLEMS OUTPUT DEBUG CONSOLE TERMINAL PORTS

akashrathor@Akashs-MacBook-Air code.cpp % clear

5

2 4 5 8 6

2 4 5 6 8

○ akashrathor@Akashs-MacBook-Air code.cpp %

zsh

Code code....